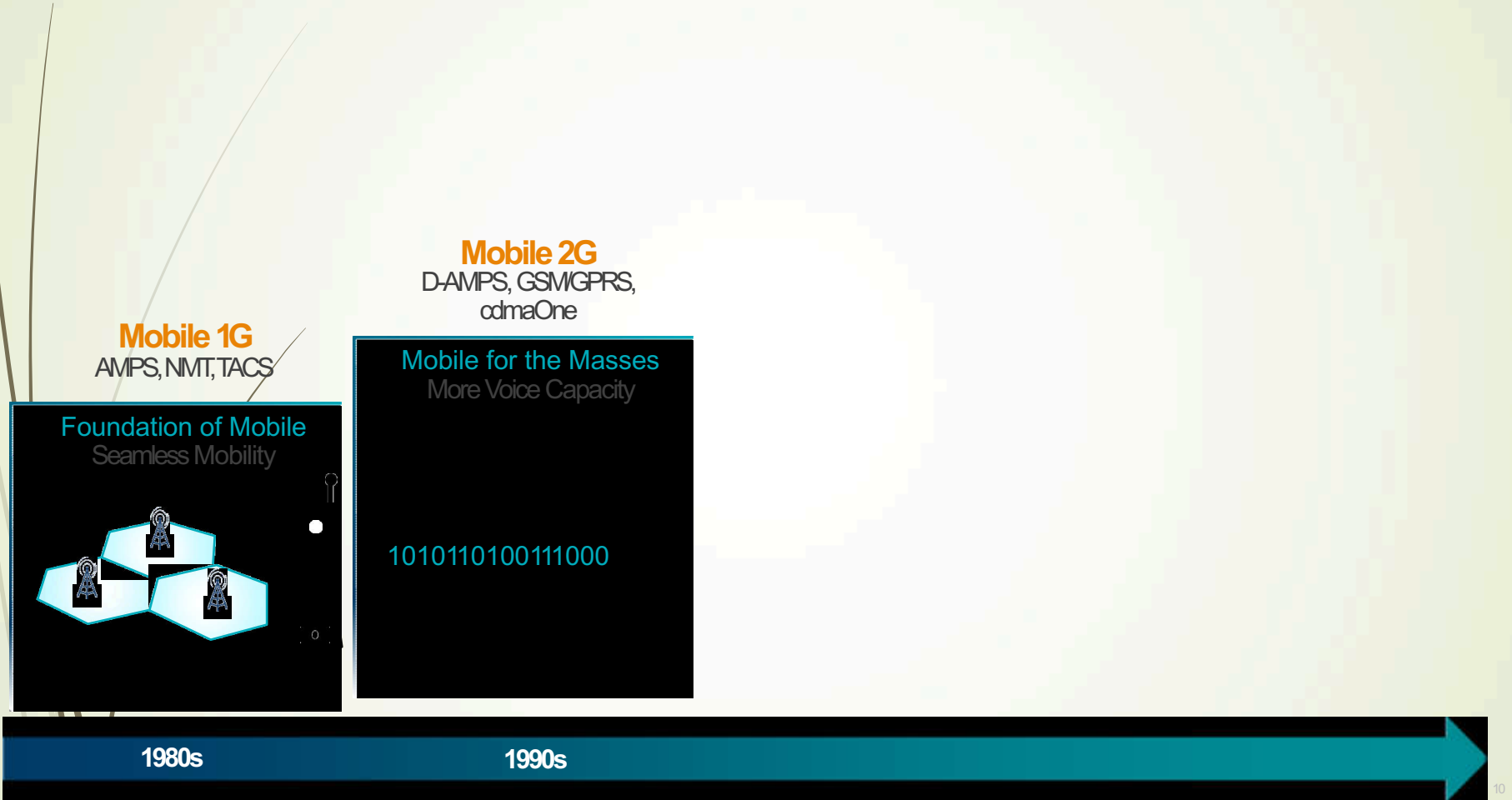


2G Offers Digital Voice Services for Higher Capacity

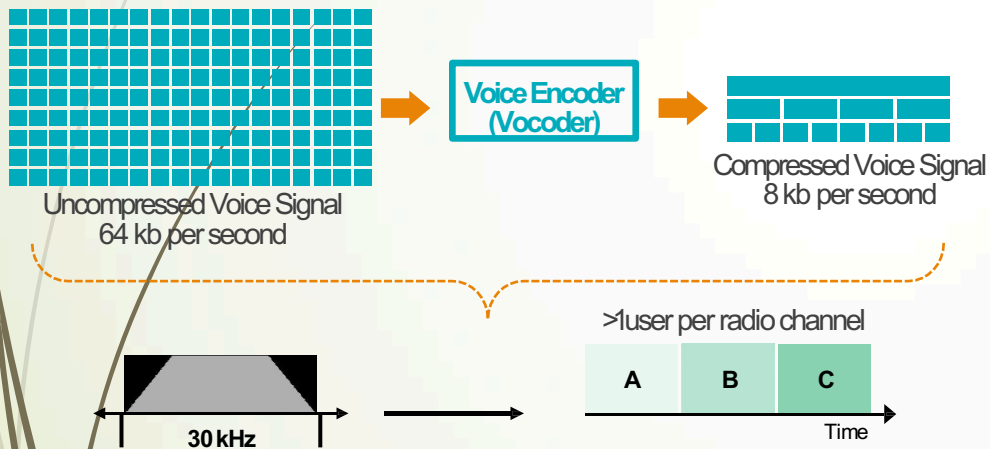


Key Innovation in 2G: TDMA

Initial 2G technologies (D-AMPS, GSM) based on TDMA

More Voice Capacity

Digital transmissions enable compressed voice and multiplexing multiple users per channel

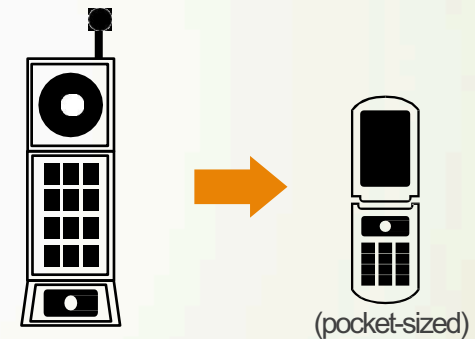


Time Division Multiple Access (TDMA)

Allows multiple users per radio channel with each user talking one at a time

Scalable Technology

Digital components cost/weight far less plus deliver more secure signal



Q: Did we abandon FDMA in 2G and beyond?

Several TDMA Standards Hit the Markets (2G)

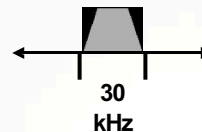
Mobile 1G(Analog) AMPS, NMT, TACS

Mobile 2G (Digital) D-AMPS

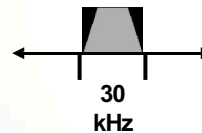
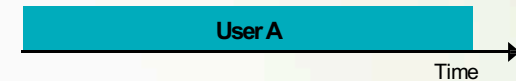
Standardized as IS-54 by TIA in 1992
Mainly in North America
No longer utilized

Mobile 2G (Digital) GSM

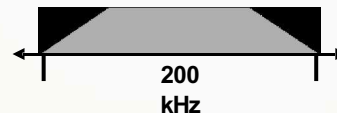
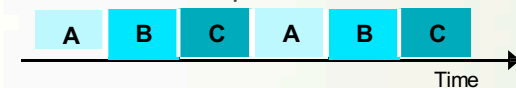
Standardized by ETSI in 1990 (phase 1)
Initiated in Europe
Still widely used today (>4B connections WW!)
Simple data services with GPRS



Only one user per radio channel



Three users per radio channel

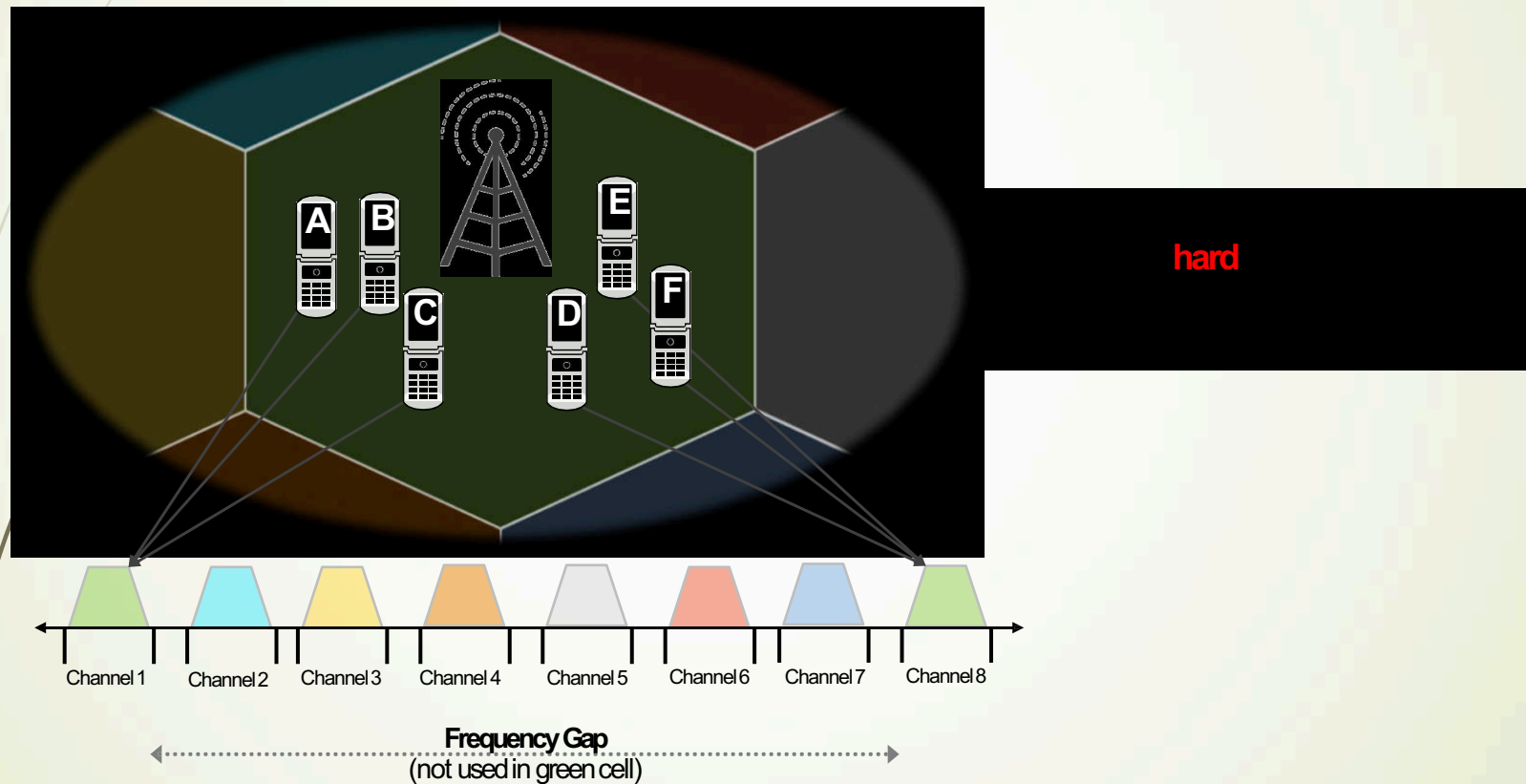


Eight users per radio channel



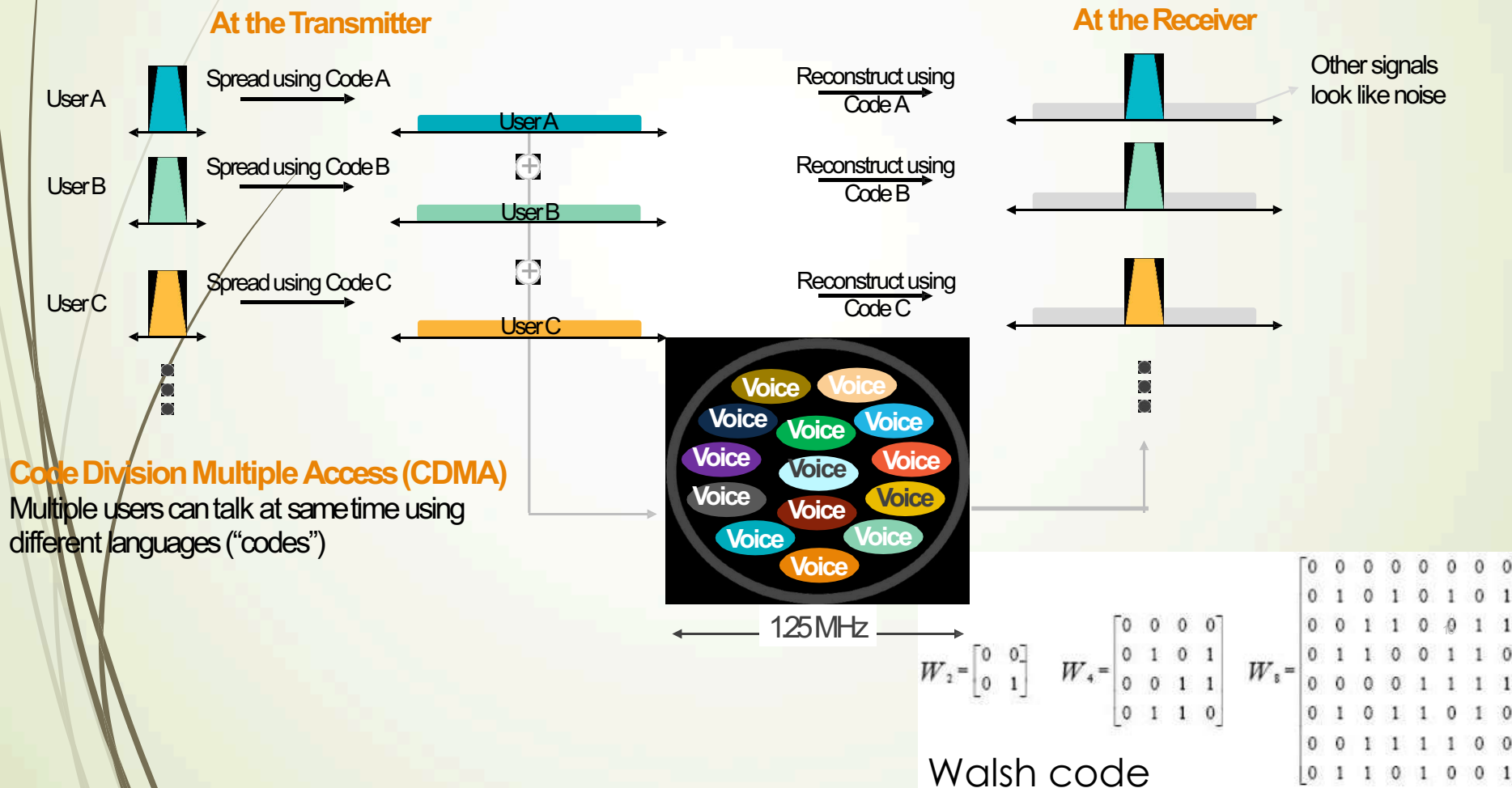
Q: Well, at the same (2G) time, Qualcomm developed IS-95, a CDMA standard. What's the key strength of CDMA?

TDMA Requires Large Frequency Gaps to Control Interference



Q: Soft handoff is another key selling point of IS-95, what is it?

CDMA Users Communicate at the Same Time



More Challenges in CDMA Networks

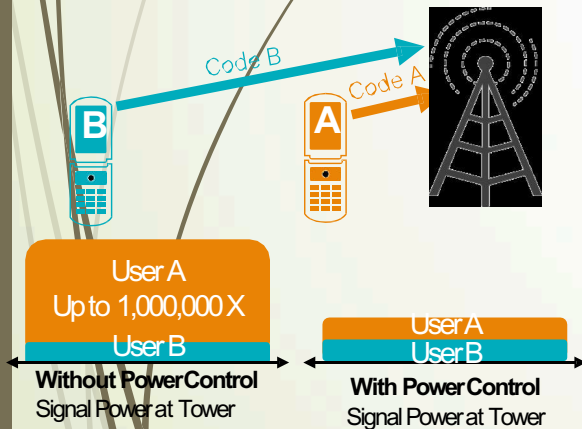
1

Near-Far Power Challenge

Users close to the tower overpower the uplink signal minimizing capacity on the shared channel

Solution:

Continuous control of transmit power based on signal strength



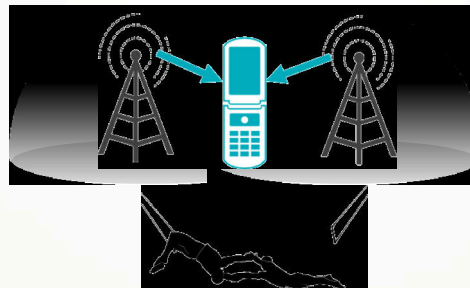
2

Cell-Edge Challenge

Interference caused by users in close proximity, on the same frequency, and communicating with different towers

Solution:

Users simultaneously communicate with multiple towers at cell edge



+ Soft (vs. Hard) Handoffs

Additional benefit of simultaneous connections – more reliable handoffs

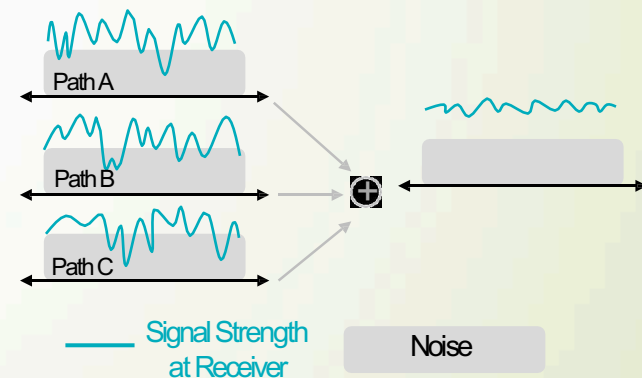
3

Multipath Fading Challenge

Interference caused by the reception of the same signal over multiple paths resulting in poor signal-to-noise ratio

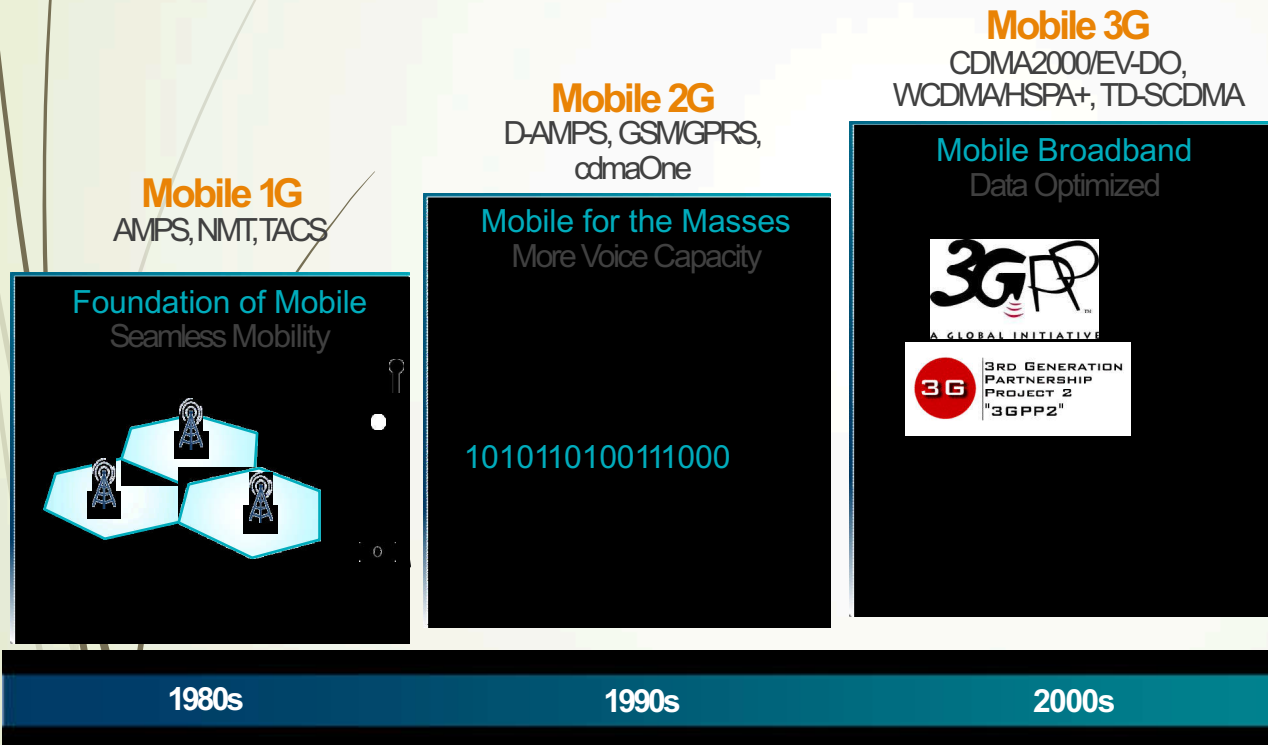
Solution:

Advanced (“rake”) receivers combine energy of multiple signal paths



3G for Mobile Data Networks

CDMA comes back in 3G networks



4G LTE Deliver Higher-Speed Data Networks

Download, browse, stream, and game faster than ever with faster and better connectivity

Connect Faster



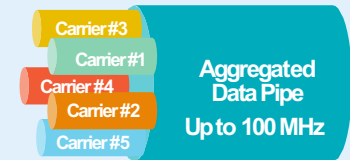
Wider Channels

Flexible support for channels up to 20 MHz enabled with OFDMA



More Antennas

Advanced MIMO techniques to create spatially separated paths; 2x2 MIMO mainstream



Carrier Aggregation

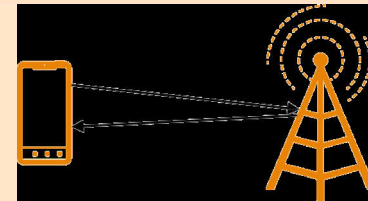
Aggregate up to 100 MHz for higher data rates – 2 carrier (2C) commercial; 3C announced¹

Connect Real-time



Simplified Core Network

All IP network with flattened architecture resulting in less equipment per transmission



Low Latencies

Optimized response times for both user and control plane improves user experience